

EL MEHDI HICHAM

Computer Vision Engineer | AI Engineer | Sports Analytics

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github.com/MehdiHCH | mehdihch.github.io/PortfolioMCH | Morocco - open to relocation & remote

PROFILE

Computer Vision Engineer specialised in **real-time video intelligence**, **multi-object tracking**, **geometric vision** and sports analytics. Experienced in delivering end-to-end systems across detection, tracking, ReID, homography, OCR, pose estimation and GPU-accelerated inference. Recent client work spans tracking-data analytics, camera calibration, Veo workflows and technical advisory.

EXPERIENCE

Computer Vision Engineer & Sports AI Consultant

07/2026 - 09/2026

SolveXI, TechnologyOne & selected international clients - Freelance / Remote

- **SolveXI / Goran Mandic:** integrated SkillCorner tracking, events and phase-of-play data into a tactical analysis pipeline, producing coach-readable insights on team shape, width/depth, possession structure, passing options, pressing behaviour and cross-match patterns.
- **TechnologyOne / Eugene Luong:** evaluated football footage quality, player detection/tracking feasibility and camera-calibration requirements; advised on CV architecture, model adaptation, Veo-camera workflows and reliable homography/pitch mapping.
- Delivered technical advisory on system design, feasibility, model selection, deployment strategy and development planning for international clients.

Tech: Python, PyTorch, OpenCV, SkillCorner Data, Multi-Object Tracking, Homography, Tactical Analytics

Computer Vision Engineer (Contract)

05/2026 - 06/2026

Priori Sports, Inc. - San Francisco, USA (Remote)

- Developed computer vision solutions for **football spatial tracking**, including calibrated spatial projection and tracking-data processing.
- Built **ball-height and trajectory estimation** from 2D tracking and camera-calibration data, including smoothing, airborne detection and spatial correction.

Tech: Python, PyTorch, OpenCV, Homography, CUDA, TensorRT, Docker

Computer Vision Engineer (Freelance)

09/2025 - 10/2025

SportScore, Paris (Remote)

- Restructured and optimised the production inference pipeline for real-time performance, separating model loading, pre/post-processing and inference into independent stages.
- Reduced inference latency by **50%** and increased throughput to **13 FPS (+117%)** through CUDA Stream parallelisation, TensorRT FP16 and asynchronous I/O.

Tech: TensorRT FP16, CUDA Streams, ONNX, PyTorch, Python

R&D; Intern - Applied AI & Computer Vision

04/2025 - 09/2025

SportScore, Paris (Remote)

- Built an automatic frame-classification module using YOLOv11m-cls, achieving **99.95% accuracy**; implemented football action detection for passes, shots, dribbles, interceptions and goals.
- Applied **camera homography and geometric transformations** for spatial analysis and pitch mapping; delivered automated performance analytics and AI-powered microservices.
- Developed player/ball detection, tracking, ReID, team assignment, jersey OCR and pose-estimation components; mentored 4 interns and maintained code quality.

Tech: Python, PyTorch, TensorFlow, OpenCV, CUDA, TensorRT, Pandas, Docker, FastAPI

Machine Learning Intern - Research & Experimental Evaluation

10/2024 - 01/2025

SETIME Lab, Ibn Tofail University

- Designed reproducible ML pipelines on a **40GB** football dataset; conducted controlled experiments with benchmarking and ablation studies.
- Achieved **0.93 recall** using RegNet-Y + Bi-GRU; performed hyperparameter tuning and architecture exploration.

Tech: Python, PyTorch, RegNet-Y, ResNet-50, Bi-GRU, OpenCV, NumPy, Pandas

SELECTED PROJECTS

OCR System for Handwritten Tifinagh - 2024

Hybrid CRNN (CNN + RNN) for Amazigh script recognition; benchmarked against PaddleOCR.

Autonomous Driving - Deep Reinforcement Learning - 2024

ROS2 + Gazebo simulation pipeline using camera and LiDAR; TD3 agent for navigation and obstacle avoidance.

SKILLS

Computer Vision: YOLOv8/11, ByteTrack, DeepSORT, OSNet ReID, OCR, pose estimation, segmentation, homography, camera calibration, pitch mapping, 3D geometry

Sports AI & Video Analytics: player/ball tracking, tracking-data analysis, tactical analytics, pitch-coordinate reconstruction, SkillCorner data, Veo-camera workflows

Performance & Deployment: CUDA Streams, TensorRT FP16/INT8, ONNX, real-time inference, Docker, AWS (S3, EC2)

Programming: Python (advanced), C/C++, Shell, SQL, FastAPI, REST API

Languages: Arabic (native), French (professional), English (professional working proficiency)

EDUCATION

Master's Degree in Artificial Intelligence - *Ibn Tofail University, Kenitra 2023 - 2025*

Bachelor's Degree in Physics (Theoretical & Mathematical) - *Ibn Tofail University, Kenitra 2020 - 2023*